

mdstats 0.20.183a0

SIZE-FIDELITY1 calibration authority

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This release implements the **SIZE-FIDELITY1** scientific calibration machinery required after the SIZE-HALVE1 target-size correction. It intentionally does not promote the three-epoch screen to certified production authority without real exhaustive MACE evidence.

Added

- **SizeFidelityCalibrationPolicy** with at least three frozen optimizer seeds, candidate coarse endpoints 3/4/5, monitor sizes 128/256/512/1024, and early equivalence widths 1/2/4 meV/Angstrom.
- **SizeFidelityExecutionPlan**, which freezes every calibration seed x hard-coverage-qualified target-size run and required checkpoint endpoints before training.
- **SizeFidelityMetric**, **SizeFidelityCandidateAssessment**, and **SizeFidelityQualificationReport** with canonical SHA-256-backed serialization and fail-closed validation.
- Public execution-plan, qualification-builder, and fail-closed qualification-validator APIs.
- Deterministic Spearman rank-correlation diagnostics for coarse-versus-final target scores. Correlation remains diagnostic and receives no authority to waive a survivor-recall failure.

Scientific acceptance

A candidate low-fidelity rule must retain **both** eventual 30-epoch target finalists through the coarse and epoch-10 screens for every frozen calibration seed. The monitor-based coarse promotion set must exactly equal the corresponding full-development promotion set, and the largest ladder boundary may not be falsely eliminated when it becomes a final target finalist.

The recommendation order is earliest faithful coarse endpoint, smallest exact monitor, then smallest tested early equivalence width at or above the current production default.

Calibration efficiency

Every checkpoint is inferred once on the complete leakage-safe development role. Candidate monitor metrics are deterministic reductions of the same authenticated per-frame prediction authority. Testing four monitor sizes therefore does not purchase four additional MACE inference passes.

With N_K hard-coverage sizes and N_s calibration seeds, the v1 calibration plan requires $N_K N_s$ uninterrupted 30-epoch runs and $5N_K N_s$ full-role inference endpoints at epochs 3, 4, 5, 10, and 30.

Qualification boundary

The supplied environment has no authorizing foundation checkpoint and no qualifying GPU campaign. This release qualifies the calibration machinery and execution contract on CPU, but the **SIZE-FIDELITY1 scientific gate remains open**. PERF-P2R remains blocked until exhaustive real-trajectory evidence produces a passing report.

Documentation

The canonical MLFF architecture advances to revision 49 and dependency-graph schema 31. The new normative specification is docs/specs/training_data/mlff_size_fidelity1_coarse_screen_calibration_spec.md.